

Wearables at Scale:

Technical progress and practical barriers for measuring physical activity in national population health surveillance

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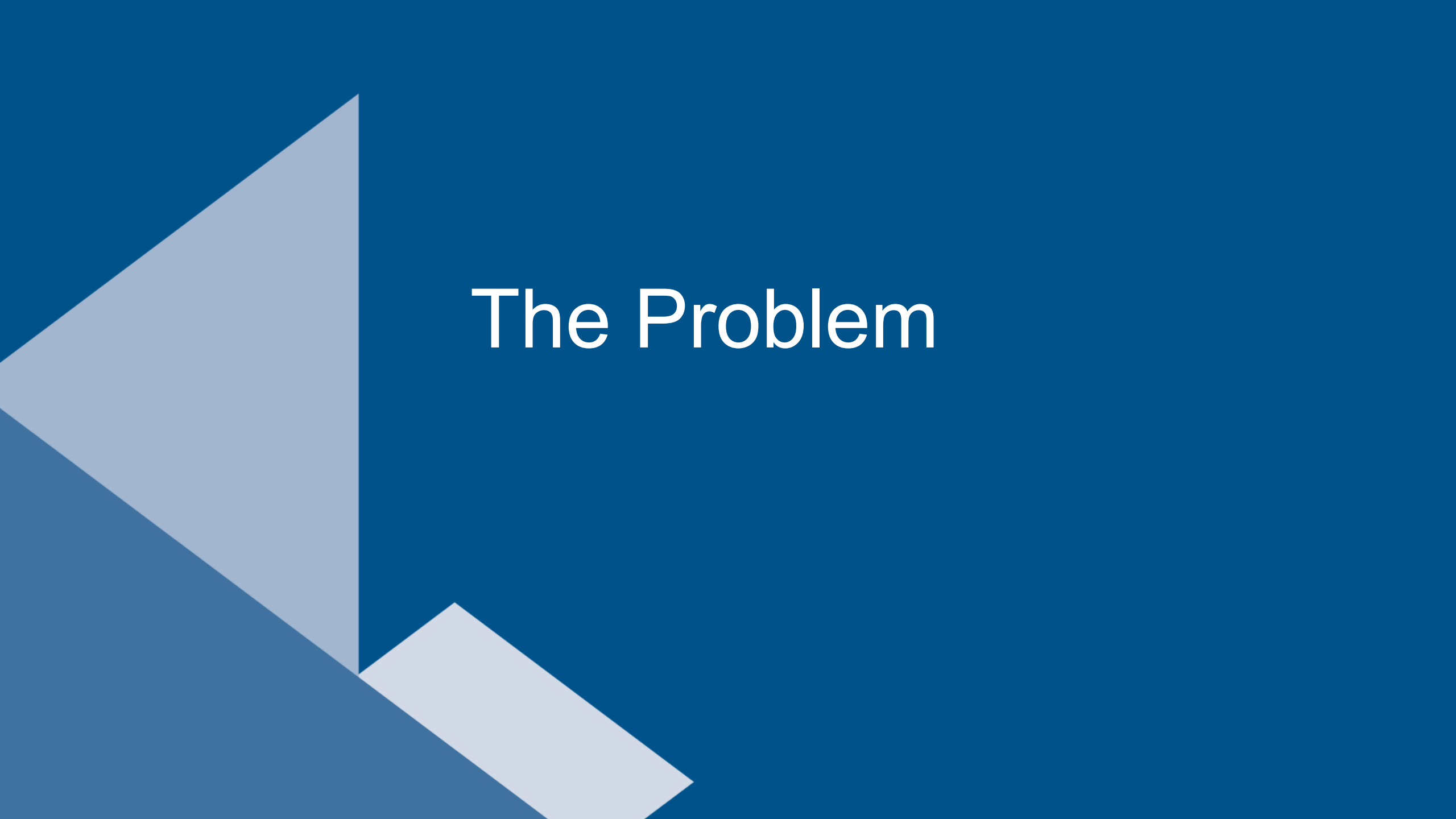
Geneva, Switzerland



13 February 2026

Health Research from Home Connect, London, U.K.





The Problem



WHO: Core functions and responsibility

- Provide leadership on global health matters
- **Monitor and assess health trends (diseases, determinants, policy)**
- **Set global norms and standards (including guidelines & guidance)**
- Articulate evidence-based policy options
- Provide technical support to countries
- **Shape the health research agenda**

So what is population surveillance?

Public health surveillance is the ongoing, systematic collection, analysis and interpretation of data with prompt dissemination of results to those who need to know, particularly those who are in a position to take action.

- Monitors health status, disease and associated risk factors in the whole population
- Provides estimates and track trends over time
- Inform health care planning and policy development
- Enables evidence-based decision making (and resource allocation!)
- It measures both the need for interventions and the effect of interventions
- It is a core public health function

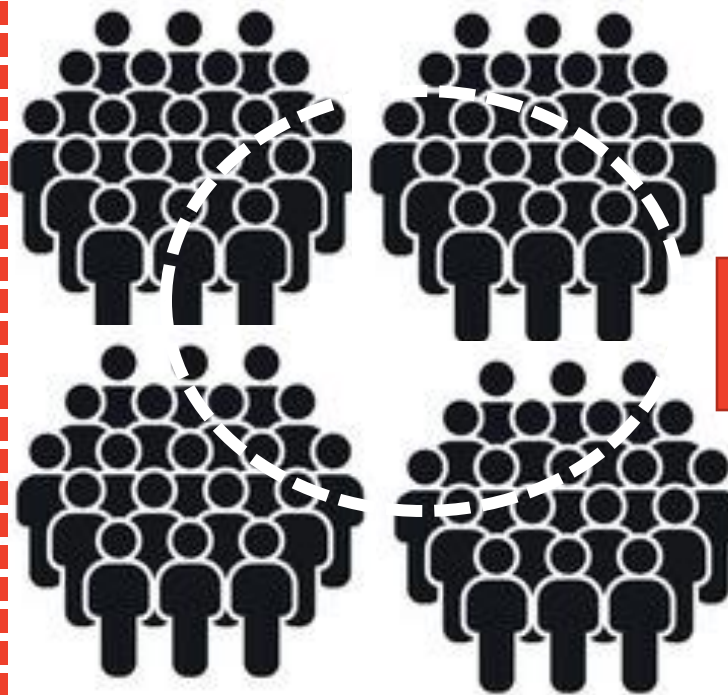
Measuring physical activity at different scales

Individual-level
measurement
focuses on a
single person



versus

Population-level measurement uses aggregate data from a large group to understand trends and overall characteristics.



Randomly
selected
representative
sample



Global monitoring - global comparisons



194 countries



What are we trying to
measure in population
health surveillance?

Global Action Plan On Physical Activity 2018-2030

(Adopted in World Health Assembly Resolution WHA71.6)



There are many ways to be active – walking, cycling, sport, active recreation, dance and play - and many policy opportunities to increase participation.

GLOBAL TARGETS

BY 2025
10%

BY 2030
15%

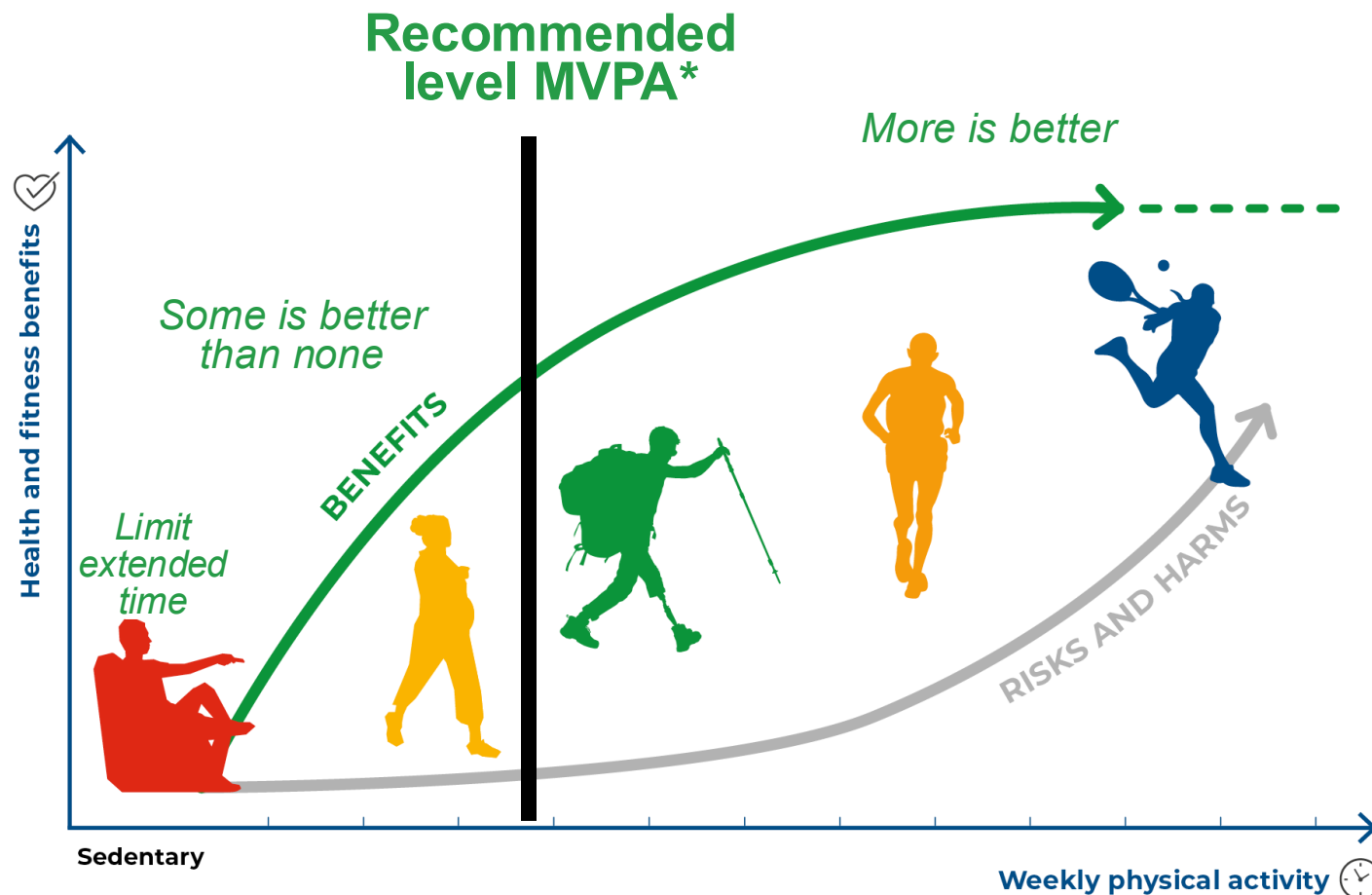
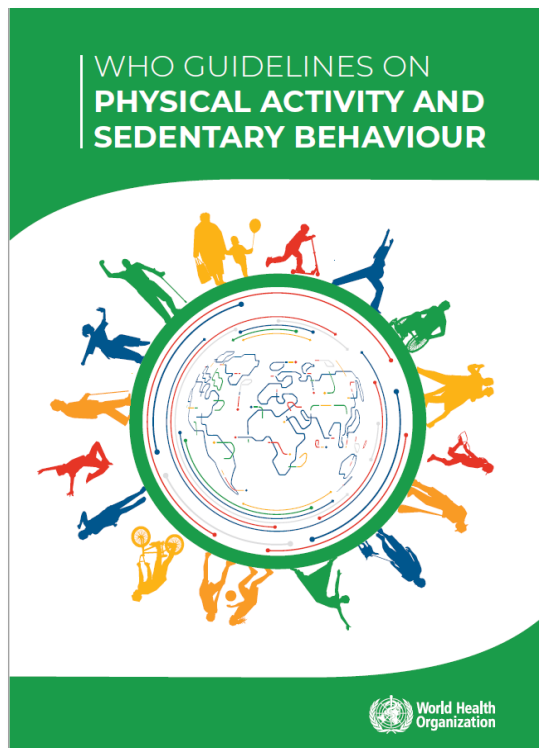
And to be more precise.....

A 15% relative reduction in the global prevalence of **insufficient physical activity** in adults and adolescents by 2030

Where....

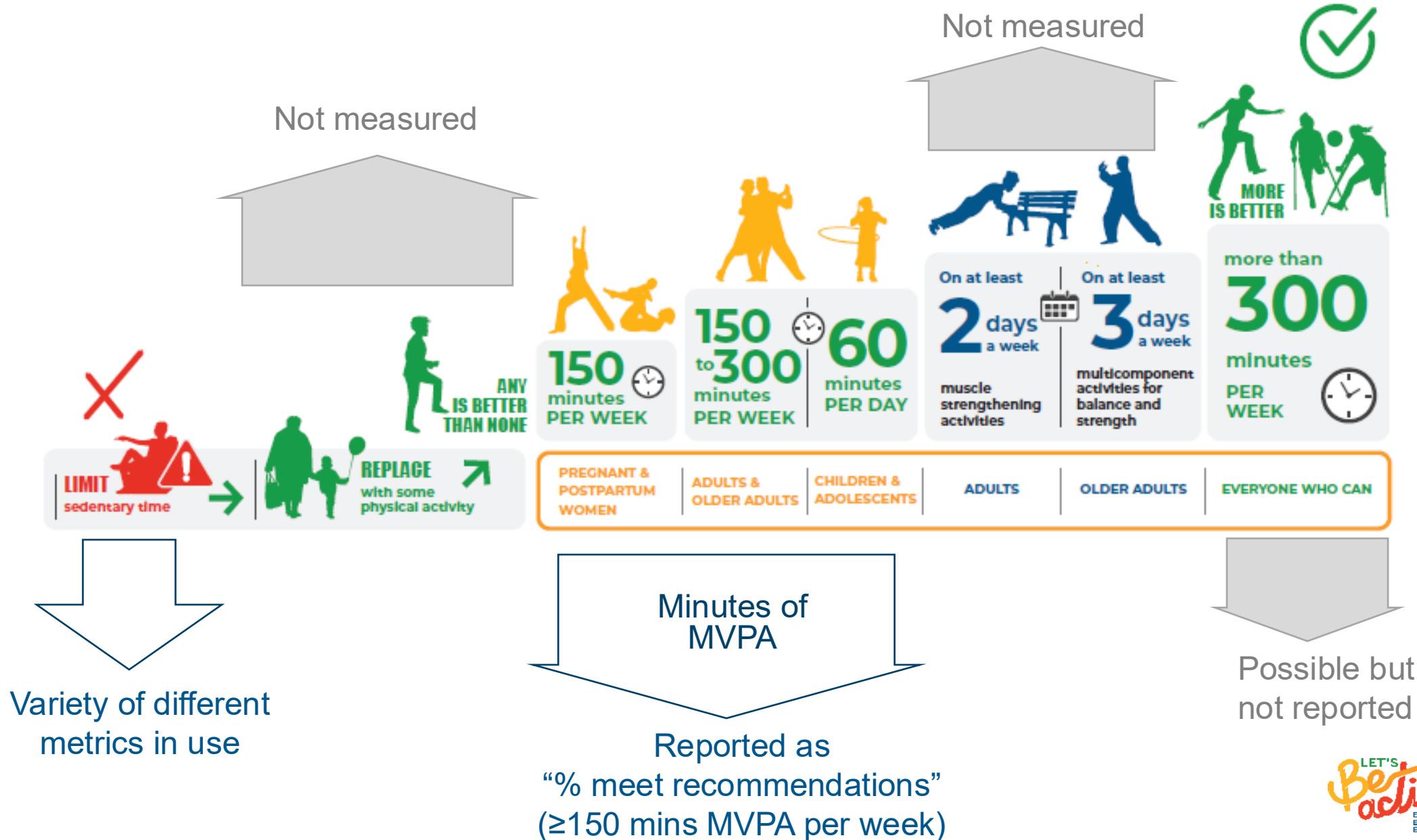
insufficient physical activity = not meeting global recommendation

Global recommendations on physical activity (adults)

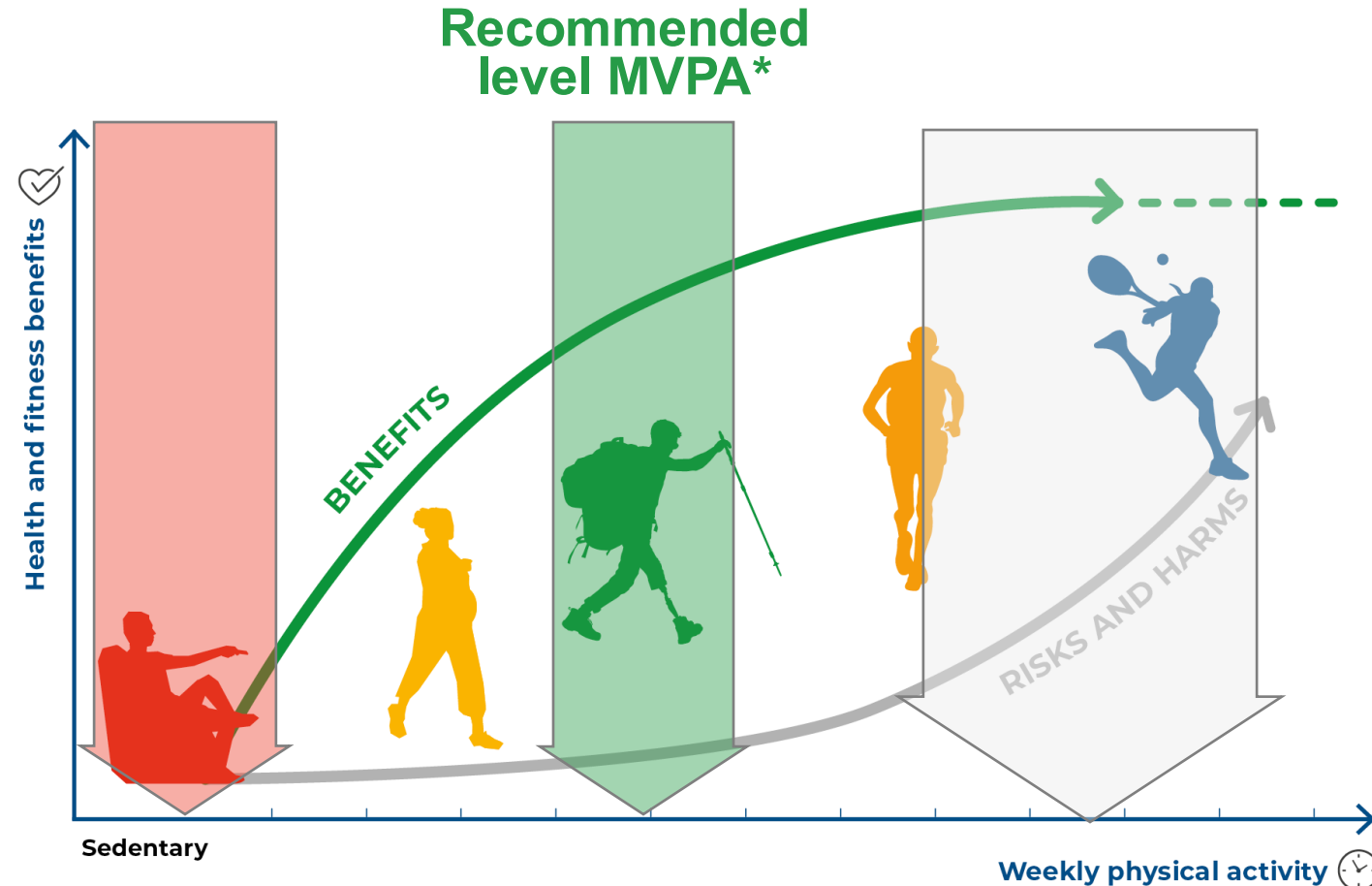
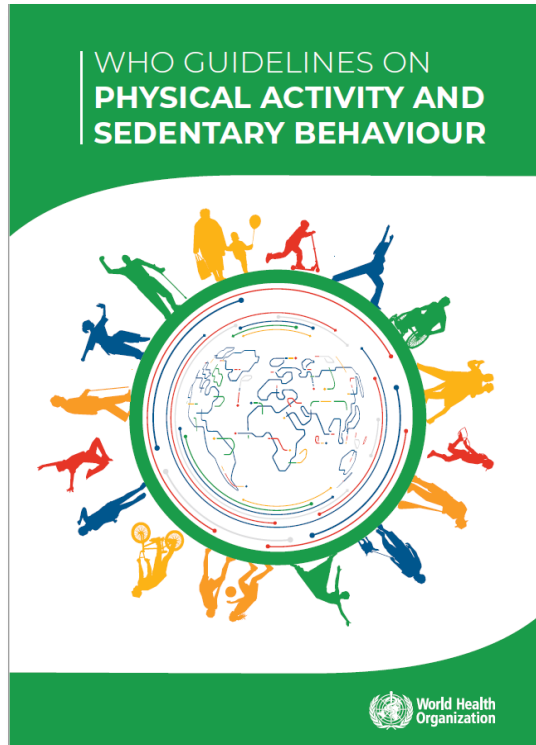


* Moderate- to vigorous- intensity physical activity

What we need to measure ?



Global recommendations on physical activity (adults)



* Moderate- to vigorous- intensity physical activity



What instruments are
we using now?

Self report questionnaire: Example GPAQ

Global Physical Activity Questionnaire (GPAQ)



WHO STEPwise approach to NCD risk factor surveillance

Surveillance and Population-Based Prevention
Prevention of Noncommunicable Diseases Department
World Health Organization
20 Avenue Appia, 1211 Geneva 27, Switzerland
For further information: www.who.int/chp/steps

GPAQ

Physical Activity

Next I am going to ask you about the time you spend doing different types of physical activity in a typical week. Please answer these questions even if you do not consider yourself to be a physically active person.

Think first about the time you spend doing work. Think of work as the things that you have to do such as paid or unpaid work, study/training, household chores, harvesting food/crops, fishing or hunting for food, seeking employment. (Insert other examples if needed). In answering the following questions 'vigorous-intensity activities' are activities that require hard physical effort and cause large increases in breathing or heart rate. 'Moderate-intensity activities' are activities that require moderate physical effort and cause small increases in breathing or heart rate.

Questions	Response	Code
Activity at work		
1 Does your work involve vigorous-intensity activity that causes large increases in breathing or heart rate like [carrying or lifting heavy loads, digging or construction work] for at least 10 minutes continuously? (INSERT EXAMPLES) (USE SHOWCARD)	Yes 1 No 2 If No, go to P 4	P1
2 In a typical week, on how many days do you do vigorous-intensity activities as part of your work?	Number of days: ____	P2
3 How much time do you spend doing vigorous-intensity activities at work on a typical day?	Hours: minutes: ____ : ____ hrs mins	P3 (a-b)
4 Does your work involve moderate-intensity activity that causes small increases in breathing or heart rate such as brisk walking (or carrying light loads) for at least 10 minutes continuously? (INSERT EXAMPLES) (USE SHOWCARD)	Yes 1 No 2 If No, go to P 7	P4
5 In a typical week, on how many days do you do moderate-intensity activities as part of your work?	Number of days: ____	P5
6 How much time do you spend doing moderate-intensity activities at work on a typical day?	Hours: minutes: ____ : ____ hrs mins	P6 (a-b)
Travel to and from places		
The next questions exclude the physical activities at work that you have already mentioned. Now I would like to ask you about the usual way you travel to and from places. For example to work, for shopping, to market, to place of worship. (Insert other examples if needed)		
7 Do you walk or use a bicycle (pedal cycle) for at least 10 minutes continuously to get to and from places?	Yes 1 No 2 If No, go to P 10	P7
8 In a typical week, on how many days do you walk or bicycle for at least 10 minutes continuously to get to and from places?	Number of days: ____	P8
9 How much time do you spend walking or bicycling for travel on a typical day?	Hours: minutes: ____ : ____ hrs mins	P9 (a-b)
Recreational activities		
The next questions exclude the work and transport activities that you have already mentioned. Now I would like to ask you about sports, fitness and recreational activities (leisure). (Insert relevant terms)		
10 Do you do any vigorous-intensity sports, fitness or recreational (leisure) activities that cause large increases in breathing or heart rate like [running or football] for at least 10 minutes continuously? (INSERT EXAMPLES) (USE SHOWCARD)	Yes 1 No 2 If No, go to P 13	P10
11 In a typical week, on how many days do you do vigorous-intensity sports, fitness or recreational (leisure) activities?	Number of days: ____	P11
12 How much time do you spend doing vigorous-intensity sports, fitness or recreational activities on a typical day?	Hours: minutes: ____ : ____ hrs mins	P12 (a-b)

- Too many questions - takes too long
- Feels repetitive to responder (and the interviewer)
- Confusion about terms
- Show card examples are useful but also limited
- Difficulty estimating time spent - leads to rounding
- Weak to moderate validity compared with device

Deaths potentially averted by small changes in physical activity and sedentary time: an individual participant data meta-analysis of prospective cohort studies

Ulf Ekelund, Jakob Tarp, Ding Ding, Miguel Adriano Sanchez-Lastra, Knut Eirik Dalene, Sigmund A Anderssen, Jostein Steene-Johannessen, Bjørge H Hansen, Bente Morseth, Laila A Hopstock, Edvard Sagelv, Peter Nordström, Anna Nordström, Maria Hagströmer, Ing-Mari Dohrn, Keith M Diaz, Steven Hooker, Virginia J Howard, I-Min Lee, Morten W Fagerland

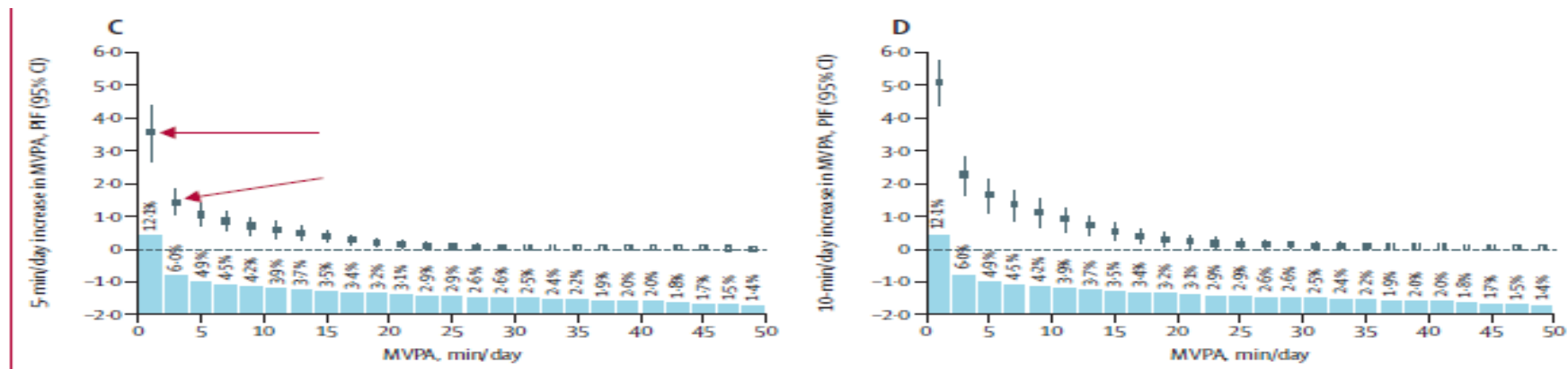


Figure 1: MVPA and mortality risk in the accelerometer consortium

Figure shows results of meta-analysis of seven studies. HRs for mortality for 5-min/day (eg, the HR for a 5-min/day increase from 1 min/day to 6 min/day [red arrow]; A) and 10-min/day (eg, the HR for a 10-min/day increase from 1 min/day to 11 min/day [red arrow]; B) increases in MVPA from observed level. Percentage PIF for 5-min/day (eg, a 5-min/day increase in MVPA in those who accumulated ≤ 2 min/day of MVPA was associated with 3.6% of preventable deaths [red arrow], and a 5-min increase among those accumulating 2–4 min of MVPA per day was associated with 1.4% of preventable deaths [red arrow]; C) and 10-min/day increase in MVPA from observed level (D). Bar charts show distribution and percentages of participants. All results are based on fully adjusted models (age, sex, accelerometer wear-time, BMI, smoking, education, cardiovascular disease, cancer, and diabetes). HR=hazard ratio. MVPA=moderate-to-vigorous-intensity physical activity. PIF=population impact fraction.

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www.thelancet.com

“Small and realistic increases in [physical activity] of 5 min/day might prevent up to 6% of all deaths.”

See Article page 339

Potential impact of wearables on physical activity guidelines and interventions: opportunities and challenges

Jason MR Gill ¹, Timothy J Chico ², Aiden Doherty,³ Jessilyn Dunn,⁴ Ulf Ekelund,^{5,6} Peter T Katzmarzyk ⁷, Karen Milton ⁸, Marie H Murphy,⁹ Emmanuel Stamatakis ¹⁰

Hundreds of millions of people own wearable devices capable of tracking their movement patterns.¹ Accelerometers are also increasingly the preferred tool to measure physical activity in research studies.² However, national and international physical activity guidelines, which recommend adults undertake at least

dose–response relationship between physical activity and health.

Recent prospective studies using wearable devices have started to transform our understanding of the association between physical activity and health outcomes. These suggest that the dose–response relationship between phys-

continue to facilitate, new insights into how both dose and pattern of physical activity affect health outcomes, for example, demonstrating potential benefits of micropatterns of VPA,⁴ and of various sedentary behaviours. The efficacy of novel interventions for the promotion of physical activity and health outcomes is most in need of research.

RESEARCH WEARABLES Most of the response behaviour

Journey of a thousand miles: from ‘Manpo-Kei’ to the first steps-based physical activity recommendations

Emmanuel Stamatakis ^{1,2}, Matthew Ahmadi ^{1,2}, Marie H Murphy ³, Timothy James Chico ⁴, Karen Milton ⁵, Borja Del Pozo Cruz ^{6,7}, Peter T Katzmarzyk ⁸, I-Min Lee ^{9,10}, Jason Gill ¹¹

Stepping—encompassing walking, running and stair-climbing—is the fundamental mode of human movement. Higher stepping volume and intensity is associated with favourable health outcomes.^{1,2} Over the last quarter of the century, stepping

challenges surrounding the addition of stepping-based recommendations to future guidelines.

DAILY STEPS: AN OLD-NEW TARGET? Current physical activity recommendations are based on weekly duration (time) of moderate and vigorous activity (MVPA).⁴ For some people steps may be an easier to

CHALLENGES Evidence synthesis Critically, guideline developers should acknowledge that the questionnaires used to derive current time-based MVPA targets⁴ and step-counting devices capture different constructs: questionnaires capture continuous blocks of time when bouts of MVPA (including stepping) occur, whereas devices capture stepping of any duration and intensity. On its own, the number of steps is an intensity-agnostic metric which does not reflect short and intermittent (up to 1–2 min) vigorous intensity bouts⁸ which may be associated with health benefits over and above stepping volume.^{1,2} Some recent 24-hour accelerometry studies using wrist or thigh-worn devices^{1,2,7,9} suggested that optimal mortality and chronic disease risk factors reduction occur at around the popular 10000 steps per day target. In its entirety, however, literature on minimal and optimal numbers of steps is far from conclusive as there are major differences in the calculation and reporting methods across studies.^{1,2,7–10}

Reporting accelerometer methods in physical activity intervention studies: a systematic review and recommendations for authors

Alexander H K Montoye,^{1,2} Robert Korycinski,⁴ Karin A Pfeiffer

Gabrys L¹, Vogt L²

ABSTRACT **Objective** This systematic review assessed completeness of accelerometer reporting in activity (PA) intervention studies and assessed related to accelerometer reporting. **Design** The PubMed database was used to identify manuscripts for inclusion. Included studies were interventions that used accelerometers, were English and were conducted between 1 Jan and 31 July 2014. 195 manuscripts from PA interventions that used accelerometers to measure physical activity were included. Manuscript completeness was assessed using 12 questions focused on 3 accelerometer areas: accelerometer information, data processing and interpretation and protocol non-compliance. **Results** Publication year, journal focus and

Accelerometry and Wearable Devices in Sports Medicine and Health Promotion

Akzelerometrie und Wearables in der Sportmedizin und Gesundheitsförderung

Summary

- Since more than two decades, accelerometers are used in sports medicine and health promotion to assess physical activity. The basis for this is the measurement of accelerations caused by physical movement. Accelerometers were originally developed for research purposes, but are also installed into smartphones, thus enabling a broad use in research and practice.
- Accelerometers can assess various dimensions of physical activity (e.g., duration, frequency, intensity) and monitor physical activity goals. In research practice, parameters like wearing time, body position, and cut-off values must be considered to get meaningful data. In addition to pre-processed physical activity data, many devices give access to raw acceleration data, which improves comparability between different types of devices.
- Consumer-graded wearable devices and smartphone apps for physical activity assessment and monitoring offer a great opportunity for large scale interventions and long-term behavior change.

CLINICAL REVIEW

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Gabrys L, Vogt L. Accelerometry and wearable devices in sports medicine and health promotion. *Dtsch Z Sportmed*. 2024; 75: 29–34. doi:10.5960/dzsm.2023.584

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Sports Med. 2021 November ; 51(11): 2237–2250. doi:10.1007/s40279-021-01543-5.

Establishing a Global Standard for Wearable Devices in Sport and Exercise Medicine: Perspectives from Academic and Industry Stakeholders

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Review

Sensor-Based Wearable Systems for Monitoring Human Motion and Posture: A Review

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Abstract: In recent years, marked progress has been made in wearable technology for human motion and posture recognition in the areas of assisted training, medical health, VR/AR, etc. This paper systematically reviews the status quo of wearable sensing systems for human motion capture and posture recognition from three aspects, which are monitoring indicators, sensors, and system design. In particular, it summarizes the monitoring indicators closely related to human posture changes, such as trunk, joints, and limbs, and analyzes in detail the types, numbers, locations, installation methods, and advantages and disadvantages of sensors in different monitoring systems. Finally, it is concluded that future research in this area will emphasize monitoring accuracy, data security, wearing comfort, and durability. This review provides a reference for the future development of wearable sensing systems for human motion capture.

Keywords: wearable sensors; motion capture; posture recognition; wearable systems

Published in: *Int J Behav Nutr Phys Act*
https://doi.org/10.1186/s13063-023-01543-5

REVIEW

The impact of selected methodological factors on data collection outcomes in observational studies of device-measured physical behaviour in adults: A systematic review

Richard M. Pulsford¹, Laura Brocklebank², Sally A. M. Fenton³, Esmée Bakker⁴, Gregore I. Mielke⁵, Li-Tang Tsa^{6,7}, Andrew J. Atkin^{8,9}, Danielle L. Harvey⁹, Joanna M. Blodgett¹⁰, Matthew Ahmadi¹¹, Le Wei¹¹, Alex Rowlands^{12,13,14}, Aiden Doherty¹⁵, Vegar Rangu¹⁶, Annemarie Koster¹⁷, Lauren B. Sherar¹⁸, Andreas Holtermann¹⁹, Mark Hamer^{10,20} and Emmanuel Stamatakis¹¹

Four objectives, 20 policy actions and a “whole of system” approach





ACTION 4.2

Enhance data systems and capabilities at the national and, where appropriate, subnational level, to support: regular population surveillance of physical activity and sedentary behaviour, across all ages and multiple domains; development and testing of new digital technologies to strengthen surveillance systems; development of monitoring systems of wider sociocultural and environmental determinants of physical activity; and regular multisectoral monitoring and reporting on policy implementation to ensure accountability and inform policy and practice.

PROPOSED ACTIONS FOR MEMBER STATES

4. Support and engage in partnerships to develop and test innovative and new digital technologies (including wearable devices) to strengthen surveillance of physical activity and sedentary behaviour, and their determinants, across all ages and abilities, with a focus on feasible and affordable solutions, particularly in LMICs.

PROPOSED ACTIONS FOR WHO SECRETARIAT

6. Provide global leadership on population monitoring of physical inactivity, and ensure the latest scientific evidence, methods and technologies are harnessed to inform global guidance, surveillance tools and protocols, for different subpopulations, including methods for data harmonization and use of wearable devices in all ages.

PROPOSED ACTIONS FOR STAKEHOLDERS*

12. Research development agencies and researchers should support research which aims to improve global and national surveillance on physical activity across all ages, and abilities, including testing of new technologies and wearable devices, and methodologies for harmonization of data.

Primary aim

To develop global guidance on the use of wearable devices to measure physical activity and sedentary behaviors in national population health monitoring systems in adults and adolescents

In the future....

sleep (when WHO has guidelines on sleep behaviours)

other age group (<10 years; <5 years)





Consultations and expert meetings to advance workplan

1



Nov 2023



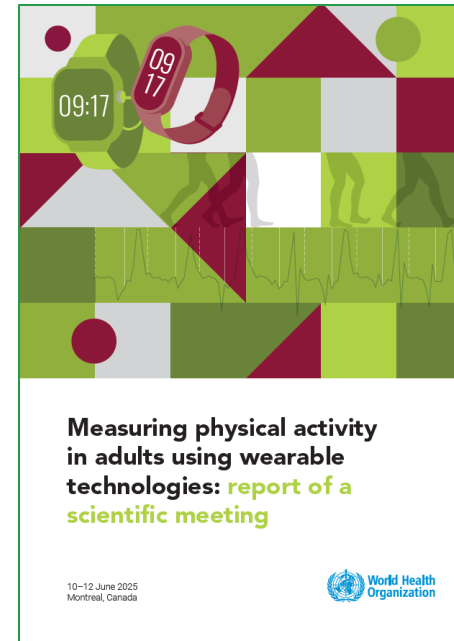
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May 2024



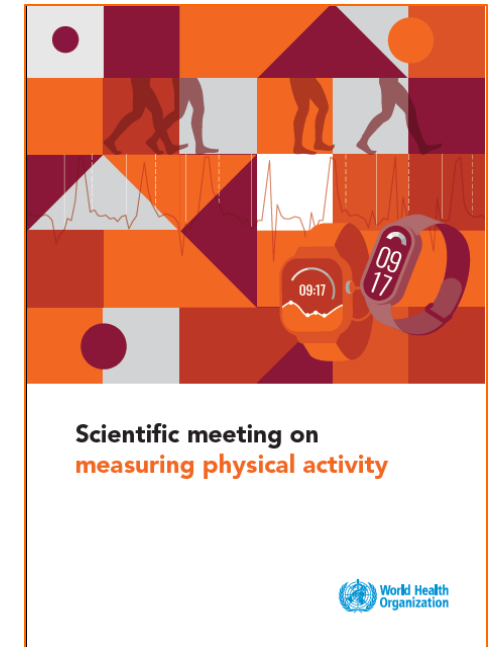
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June 2025



4



Nov 2025

4th meeting report
FORTHCOMING

Acknowledgements: To the participants, contributors and funding organizations



Consultations and expert meetings to advance workplan

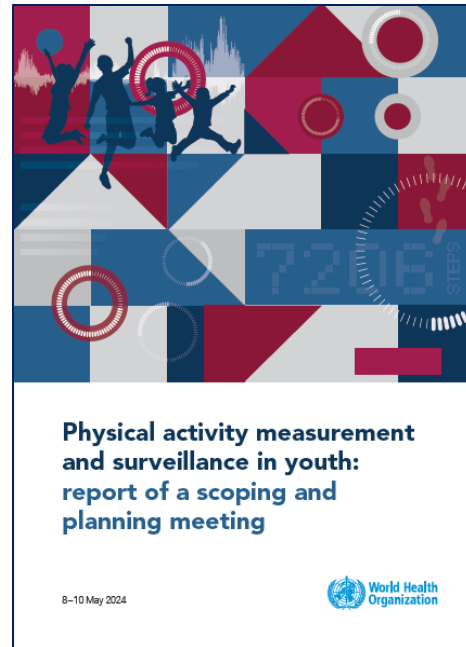
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Nov 2023



2



May 2024



- Self-report measurement has persistent and serious limitations *≠ fit for the future*
- Rapid advances in sensors, wearables and analytics = *very promising*
- Extensive use in cohort studies = *provides evidence of use at scale and need to align*
- **AND** some interest from governments
- **BUT** - uncertain methods, feasibility, cost, system integration, comparability = **need to look more!**

Acknowledgements: To the participants, contributors and funding organizations



3



**Measuring physical activity
in adults using wearable
technologies: report of a
scientific meeting**

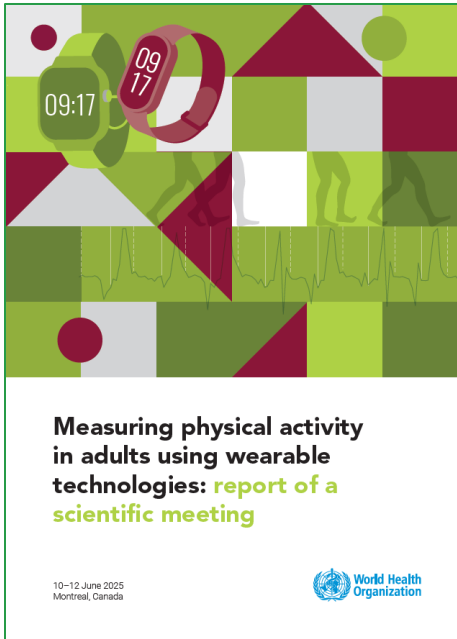
10–12 June 2025
Montreal, Canada



June 2025



3



June 2025



Meeting Objectives

1. Agree a core set of primary output metrics
2. Develop draft device specifications = required features to be “fit for purpose” (& mindful of cost)
3. Review performance of open-source algorithms - including raw data processing, validation methods, study samples
4. Discuss applicability to older adult populations

Core metrics - to focus guidance on initially

1. moderate-to-vigorous physical activity (MVPA) (mins)
2. step count – (per day)
3. sedentary behaviour (mins)

At this time:

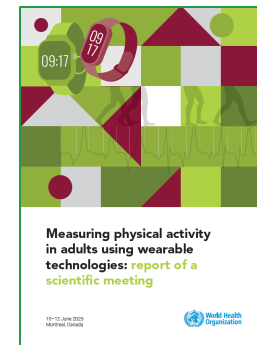
- *Sleep and muscle strengthening out of scope*

Device specifications (and protocols): *DRAFT at a glance table*

Wear protocols summary	
2.1 Wear location	Wrist worn – non-dominant AND Recommended to record the wear location for each responder as some individuals may decline to wear on non-dominant wrist
2.2 Wear time	Wear days: Minimum of 7 days / Optional: collect >7 days Wear time: No decision
2.3 Device orientation	Specification of non-dominant wrist
Device properties including sensor specifications summary	
3.1 Number of axes	Minimum: Triaxial
3.2 Dynamic range	Minimum: +/- 6g
3.3 Resolution	Minimum: 12 bits – 3 milli-g (milli-gravity units)
3.4. Sensitivity	A Least Significant Bit (LSB) of ~0.003
3.5 Noise Floor	Minimum: 1–5 milli-g
3.6 Temperature sensitivity	Minimum: operational range should be minus 20°C to plus 50°C
3.7 Sampling frequency	Minimum: 30Hz Optional – 40Hz or higher
3.8 Memory capacity	Minimum: Memory capacity must be sufficient to support data collection at the specified number of days, hours per day and sample frequency AND Raw accelerometer data must be stored in a non-volatile was (to avoid data loss should battery life end)
3.9 Battery life	Minimum: Battery needs to be sufficient to support the required data collection days and hours per day at specified sample frequency AND Preferably the device should not require the user to re-charge during the data-collection period AND If the device does require re-charging there must be mechanism(s) to protect the data and avoid user access or user download

3.10 Battery recharging and replacement	Minimum: Device must be re-usable and re-chargeable by central coordinator to enable re-use within the surveillance programme AND Devices must be re-chargeable at scale (i.e., with use of a docking station for large batch recharging)
3.11 Feedback to user	<u>Provision of feedback on physical activity/sedentary:</u> No feedback should be provided to user on physical activity (steps, minutes or other metrics) or on sedentary behaviours/time <u>Provision of other feedback or device uses:</u> No decision
3.12 Use of multi-sensor approach	Minimum: (i) Battery health and (ii) Temperature Optional – Temperature and battery health sensors can be switched on/off depending on country context. <u>OTHER SENSORS (to assist analysis/ other purposes/research)</u> (i) Heart rate (via PPG) – no decision (ii) Global positioning systems (GPS) – exclude (not required) (ii) Gyroscope – optional – as this has significant impact on battery requirements
3.12 Waterproof and dustproof and intended for prolonged skin contact	Minimum: (i) Water resistant (IPx8) (ii) Dust tight (IP6x) (iii) materials certified as hypoallergenic and biocompatible and intended for prolonged skin contact
3.13 Band fitting	Minimum: Wrist band must be adjustable, removable and cleanable or disposable (and replacement units available)
3.14 Data handling and transfer	No decision

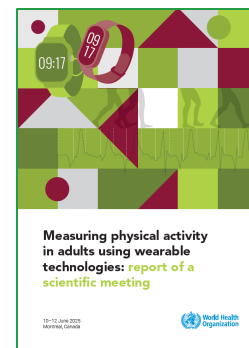
Processing protocols	
4.1 Access to raw signal	Minimum: Raw accelerometer data must be accessible
4.2 Frequency filtering	Minimum: 20 Hz Butterworth filter
4.3 Epoch length	Maximum: 60 seconds Optional 10 sec or 30 seconds – not decided
4.4 Minimum bout length and interruption tolerance	<u>Use of minimum length of time (i.e. defining a "bout"):</u> Not required/recommended to use in data analysis. <u>Tolerance for interruptions:</u> No decisions
4.5 Identifying non-wear	No decisions
4.6 Minimum wear-time inclusion criteria	No decisions



Annex: Provides detailed tables with definition and rationale

Table A3.3. Issues related to device properties (overarching recognition that decisions on these issues have implications on battery life)

Category	Issue	Definition	Explanation of important and effect on estimate of physical activity	Summary of meeting recommendations on specifications
3. Device properties including sensor specifications	3.1 Number of axes	The number of axes (between 1 and 3) on which the device measures acceleration	A single axis has less sensitivity to movement compared to three axes – most devices now use triaxial sensors.	Minimum: Triaxial
	3.2 Dynamic range	The range of acceleration that can be measured	The maximum measurable acceleration without clipping. The range set must be able to capture the majority of human movement which occurs between a range of 1–2 g (as example for Actigraph GT9x it is +/- 8g).	Minimum: +/- 6 g <i>Rationale</i> - +/- 6 g is considered to be an adequate buffer given most human movement is between 1–2 g and at 12 bits the resolution (i.e. sensitivity) has a Least Significant Bit (LSB) of 0.003 g (that is 12g/4096) and this could be considered “good enough” for physical activity recognition.
	3.3 Resolution	The smallest change in acceleration that the device can detect	The resolution is determined by the number of bits used in the analogue-to-digital conversion (ADC) process.	Minimum: 12 bits <i>Rationale</i> Selected to balance competing demands, noting that higher resolution will require more power and that more bits require greater storage capacity.
	3.4 Sensitivity	The output of the accelerometer per unit of acceleration (typically measured in g, where 1g = 9.81m/s ²)	Higher sensitivity indicates that the sensor can detect smaller changes in acceleration. Manufacturers typically state the specification and conduct laboratory testing. Independent calibration testing can be conducted to ensure output corresponds to known acceleration level.	As sensitivity is a function of the dynamic range and resolution – using proposed +/-6 g and with 12 bits – the the resolution (sensitivity) is at a Least Significant Bit (LSB) of ~0.003.



4



**Scientific meeting on
measuring physical activity**



Nov 2025

*4th meeting report
FORTHCOMING*



4



Nov 2025

4th meeting report
FORTHCOMING

Meeting Objectives

1. Review country experiences within national systems
2. Examine evidence on wrist-worn measurement of cycling
3. Identify priority actions for integration into STEPS and national systems
4. Revisit standardization - what is needed and what is not?

4



Nov 2025

4th meeting report
FORTHCOMING



What did we learn?

Different national models, different aims

- Estimating prevalence
- Understanding patterns of movement
- Gaining insights to inform policy and behaviour change interventions
- Use wearable
 - in parallel to questionnaire
 - as replacement to questionnaire

= Tension is not “device vs self-report” but how to combine them effectively

Representativeness = dominant challenge

- Who is typically missed?
 - Older adults
 - Lower socioeconomic groups
 - Migrants
 - People with poorer health/functional limitations

= bias becomes a health equity issue
- Risk of bias even with large sample size
- Shared experiences but effective mitigation strategies remains limited

What did we learn?

Differences in methods

- Device choice
- Wear location
- Protocols
- Analyses

- And much more.....

SUMMARY TABLE 1 COUNTRY SYSTEM/STUDY OVERVIEW

	1.1 SYSTEM / STUDY NAME	3.1 DEVICE	3.4 WEAR LOCATION	1.6 RECENT YR	1.5 FIRST YR	1.7 # TIMES	1.4 Partners	3.3 Device Procurement	4.2 (sub)Sample (initial ¹)
Canada	Canadian Health Measures Survey (CHMS)	Actigraph wGT3x-BT	Waist/hip	2025-27	2007	8 cycles	None	purchased	3,264
Denmark 1	Danish National Health Survey (DNHS)	SENSmotion	Thigh	2023	2023	1	Express/AS Sensmotion	Purchased	6,993
Denmark 2	Moving Denmark 2020	Axivity AX3	Thigh	2021/22	2021/22	1	University	Purchased	1,248
Finland 1	Healthy Finland 2022-23	Actigraph GT9X Link	Wrist, ND*	2023	2017	2	University	Purchased	1,980
Finland 2	FINFIT STUDY	UKK- RM42	Waist, ND*+ Wrist at night	2025-26	2017-18	3	None	Own device	10,500
Japan	National Health and Nutrition Survey (NHNS-J)	Yamax (pedometer)	Waist/hip (front pocket from 2023)	2024	1989	34	Yamasa Corp NIBN	Purchased	4,272
Norway	Norwegian National PA Surveillance (NNPAS)	Actigraph GTX3X+BT	Waist/hip	2022/22	2008/09	3	Ministry & University & NIPH	Purchased	6,089
Sweden	Health on Equal Terms (HLV)	UKK - RM42	Waist, ND*+ Wrist at night	2025	2018	2	Ministry&UKK + others	Borrowed	4,000
USA	National Health & Nutrition Examination Surv (NHANES)	Actigraph GT3x	Wrist – ND*	2011/14	2003/06	2	NIH, CDC	Purchased	16,000
Netherlands	Accelerometer pilot	ActivPAL 3/4	Thigh	2024	2023	pilot	RIVM, CBS, ++	Purchased	1,100

1 Actual device (waist worn) used for cycles 1- 6 – changed to Actigraph in cycle 7 in 2022 *ND non dominant wrist ¹ rough indication of initial/starting device sample

SUMMARY TABLE 3 REQUIREMENTS AND TRAINING (AND INDICATIVE SAMPLE SIZE)

	1.1 SYSTEM / STUDY NAME	3.1 DEVICE	2.8 Wear Time	6.11 Recharge required	3.5 Other sensors?	5.5. Device problems %	6.2 Data download	8.1 Admin/Data Training?
Canada	Canadian Health Measures Survey (CHMS)	Actigraph wGT3x-BT	7 /24	No	Gyroscope PPG	Malfunction: 2%	StatCan Secure Server	Admin-Y Data- N
Denmark 1	Danish National Health Survey (DNHS)	SENSmotion	7 /24	No	Gyroscope	Very few	Amazon webserver	Admin-Y Data- N
Denmark 2	Moving Denmark 2020	Axivity AX3	7 /24 -RW*	No	Skin sensor	Minor – few	University server	Admin-Y Data- N
Finland 1	Healthy Finland 2022-23	Actigraph GT9X Link	7 /24 -RW*	No	No	Battery (COVID)	Ssecure server in THL	Admin-N Data- N
Finland 2	FINFIT STUDY	UKK- RM42	7 /24 -RW* + switch**	No	Y - posture metrics	None	UKK server	Admin-Y Data- N
Japan	National Health and Nutrition Survey (NHNS-J)	Yamax (pedometer)	1 Day waking hrs	No	No	No info	Manual diary data entry	Admin-Y Data- N
Norway	Norwegian National PA Surveillance (NNPAS)	Actigraph GTX3X+BT	7/waking hr only - RW*	No	No	2-3%	University server	Admin-Y Data- N
Sweden	Health on Equal Terms	UKK - RM42	7 /24 -RW* + switch**	No	Y - posture metrics	Few	University&PH agency	Admin-N Data- N
USA	NHANES	Actigraph GT3x	7 /24	No	Light sensor	Some detected on return	Contractor & NCHS	Admin-Y Data- Y
The Netherlands	Accelerometer pilot	ActivPAL 3/4	9 days / 24 RW*	Yes	Gyroscope PPG	Some battery issues	At survey center & RIVM	Admin-Y Data- N

*RW-Instructed to Remove for Water based/shower/bath: **Switch = instruction for different wear location at night:

What did we learn?

What to standardize — and what not?

- **Algorithms:** yes – critical
- **Device wear location:** a critical choice – but solutions emerging
- **Device choice:** less important
- **Wear protocols:** yes - set minimum

AND

- **Device function specifications:** yes – set minimum and optimal

What did we learn?

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- **Device function specifications:** yes – set minimum and optimal



So, what next?

- Complete global guidance - based on 'good enough'
 - Conduct demonstration studies - "trial and adapt"
 - Address knowledge gaps with targeted research – algorithm, training data, LMIC
 - Govt engagement and policy-facing communications
 - Resource mobilisation and investment
-
- A continued role for WHO in convening, policy alignment, technical stewardship, and advocacy for advancing wearable agenda in national health surveillance systems



1 Geneva, Nov 2023



3 Montreal, June 2025



2 Geneva, May 2024



4 Utrecht, Nov 2025

“Let us not let perfection get in the way of starting - it is time for this transformational leap in surveillance methods”

Meeting Delegate

Thank you